BTS-VI(R/S)-05-24-2462 Reg. No. 2 - 6 2 1 0 6 7

C

B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

SE -19-206-0605 ENVIRONMENTAL ENGINEERING AND MANAGEMENT
(2019 *Scheme*)

Time: 3 Hours Maximum Marks: 60

Course Outcomes
On successful completion of the course, the students will be able to:

CO1: Summarize the most suitable technique for air pollution monitoring and control for a given application.
CO2: Recognise the type of unit operations and unit processes involved in wastewater treatment plants.
CO3: Explain the techniques for the disposal and management of urban solid wastes and hazardous wastes.
CO4: Exemplify the tools for environmental management in industries.
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

PART A
(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Describe the various assumptions and limitations of Gaussian Plume model.	3	L2	1	2
	(b)	Explain the advantages and disadvantages of Wet Scrubbers.	3	L2	1	2
	(c)	Choose the operational problems of Activated Sludge Process and Trickling filter.	3	L3	2	2
	(d)	Identify the purpose and importance of flow equalization in wastewater treatment.	3	L3	2	2
	(e)	Describe the importance of transfer station in municipal solid waste management.	3	L2	3	3
	(f)	Identify the methods of composting solid wastes manually.	3	L3	3	3
	(g)	Summarize the various stages of environmental auditing.	3	L2	4	3
	(h)	Identify the methodology of conducting Life Cycle Assessment.	3	L3	4	3

PART B

		(4 × 12 = 48)				
II.	(a)	Explain the sources of air pollutants and its effects on man and environment.	5	L2	1	2
	(b)	Examine the ground level concentration of SO_2 at a distance 2 km downwind for the following locations. (i) Central line of the plume. (ii) At a cross-wind distance of 0.5 km from central line [Given: σ_z = 130 m, σ_y = 220 m, stack height = 80 m, Emission rate of SO_2 = 163.19 g/sec, avg. wind speed = 8 m/s]	7	L4	1	1
		OR				
III.	(a)	Explain the various types of plume behaviour from tall stacks and their effects on buildings and terrain.	6	L2	1	2
	(b)	Summarize the working of an electrostatic precipitator and cyclone separator with neat sketches.	6	L2	1	1

(P.T.O.)

			Marks	BL	CO	PO
IV.	(a)	Identify the various stages of bacterial growth with a neat sketch.	6	L3	2	1
	(b)	Explain briefly the mechanism of anaerobic treatment of wastewater using UASB.	6	L2	2	2
		OR				
V.	(a)	Identify the stoichiometry of biological nitrification and de-nitrification processes.	6	L3	2	2
	(b)	Describe the working of RBC with a neat sketch.	6	L2	2	2
VI.	(a)	Outline the sources and composition of solid wastes.	6	L2	3	2
	(b)	Explain the classification of hazardous waste based on their properties with examples. What are their effect on human and environment?	6	L2	3	2
		OR				
VII.	(a)	Describe the methods used for the collection and transportation of municipal solid waste.	6	L2	3	2
	(b)	With a neat sketch explain the Landfill method used for the treatment and disposal of hazardous waste.	6	L3	3	3
VIII.	(a)	Identify the importance of 'Cleaner Technologies' in environmental pollution prevention.	6	L3	4	2
	(b)	Explain the industrial ecology and industrial symbiosis with the help of an example.	6	L2	4	3
		OR				
IX.	(a)	Construct a flow chart showing the steps involved in conducting an Environmental Impact Assessment (EIA) for a highway project.	6	L3	4	2
	(b)	Describe the importance of public participation in environmental decision making.	6	L2	4	3

Bloom's Taxonomy Levels
L1 - 61%, L2 - 33%, L4 - 6%.

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